Chapter 9. Building and Extending Loglinear–Logit Models

Collapsibility, ordinal association, rates, sparseness

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This chapter turns the loglinear models of Chapter 8 into a working toolkit: association graphs that display conditional independence and tell us when a table may be collapsed (Section 9.1), model selection and comparison (Section 9.2), residual diagnostics (Section 9.3), models that use the ordering of categories (Section 9.4), and Poisson regression for rates (Section 9.5).

9.1 Association graphs and collapsibility

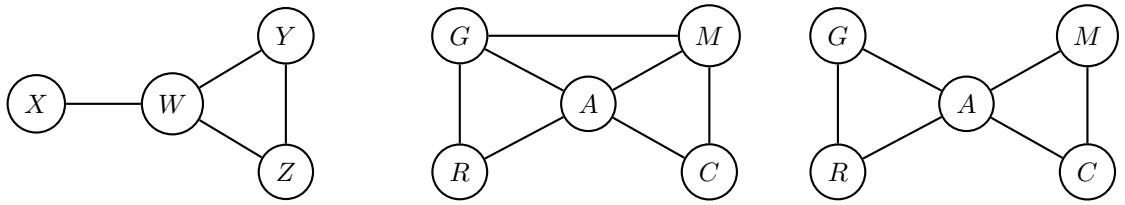
Motivation. With four or five variables it is hard to read conditional independence off a model formula. Drawing the model as a graph turns these questions into questions about paths, and it also tells us when a table can be collapsed over some variables without changing the associations of interest.

9.1.1 Association graphs

Definition (Association graph, path, separation)

The *association graph* of a loglinear model has a vertex for each variable and an edge between two variables if and only if the model contains a term involving both, i.e. if they are conditionally associated. A *path* is a sequence of edges leading from one variable to another. A set S of variables *separates* X and Y if every path between X and Y passes through a vertex of S.

Models with the same pairwise associations have the same graph; e.g. (WX, WY, WZ, YZ) and (WX, WYZ) share Figure 1.



(WX, WY, WZ, YZ) model 6: $(AC, AM, CM, AG, AR, GM, GR)$ model 7: (AC, AM, CM, AG, AR, GR)

Figure 1: Association graphs. Left: book Figure 9.1; W separates X from $\{Y, Z\}$. Middle and right: two models for the five-way student survey (Example 9.5); in model 7, A separates $\{G, R\}$ from $\{C, M\}$.

Theorem 9.1 (Separation implies conditional independence)

In a hierarchical loglinear model, if a set S of variables separates X and Y in the association graph, then X and Y are conditionally independent given S (after marginalizing over all other variables). More generally, if S separates sets A_0 and B_0, then $A_0 \perp B_0 \mid S$.

Proof. [Supplement]

Step 1. Let V be the set of all variables, A the set of vertices reachable from X by paths avoiding S, and $B = V \setminus (A \cup S)$. Since S separates X and Y, $Y \in B$.

Step 2. Every term T of the model lies in $A \cup S$ or in $B \cup S$. Indeed, if T contained $a \in A$ and $b \in B$, then the hierarchical model would contain the two-factor term $\{a, b\} \subseteq T$, so a–b would be an edge and b would be reachable from X avoiding S, i.e. $b \in A$, a contradiction.

Step 3. Grouping terms accordingly, $\mu(\mathbf{x}) = f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)$ (assign terms inside S to f). Hence, for fixed $\mathbf{x}_S$,

$$P(\mathbf{x}_A, \mathbf{x}_B \mid \mathbf{x}_S) = \frac{f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)}{\sum_{\mathbf{a}} f(\mathbf{a}, \mathbf{x}_S) \sum_{\mathbf{b}} g(\mathbf{b}, \mathbf{x}_S)},$$

which factors into a function of $\mathbf{x}_A$ times a function of $\mathbf{x}_B$: $\mathbf{x}_A \perp \mathbf{x}_B \mid \mathbf{x}_S$.

Step 4. Summing over the coordinates of A other than X and the coordinates of B other than Y preserves the product form, so $X \perp Y \mid S$. The statement for sets $A_0 \subseteq A$, $B_0 \subseteq B$ is the same computation. ■

In Figure 1 (left), W separates X and Y, so $X \perp Y \mid W$ (and also given $\{W, Z\}$); likewise $X \perp Z \mid W$.

9.1.2 Collapsibility in three-way contingency tables

Theorem 9.2 (Collapsibility conditions)

In a three-way table, the XY marginal odds ratios equal the XY conditional odds ratios (at every level of Z) if Z and X are conditionally independent given Y, or Z and Y are conditionally independent given X; that is, under models (XY, YZ) or (XY, XZ) (or simpler models).

Proof. [Supplement]

Step 1. Under (XY, XZ), by Lemma 8.5, $\mu_{ijk} = A_{ij}C_{ik}$. The conditional odds ratio at level k is

$$\frac{A_{ij}C_{ik}\, A_{i'j'}C_{i'k}}{A_{ij'}C_{ik}\, A_{i'j}C_{i'k}} = \frac{A_{ij}A_{i'j'}}{A_{ij'}A_{i'j}},$$

free of k. The marginal table is $\mu_{ij+} = A_{ij}C_{i+}$, and the row factors $C_{i+}, C_{i'+}$ cancel in the odds ratio, giving the same value.

Step 2. Under (XY, YZ), $\mu_{ijk} = A_{ij}B_{jk}$ and $\mu_{ij+} = A_{ij}B_{j+}$; now column factors cancel, with the same conclusion. ■

Example 9.3 (Collapsibility in the student survey)

In (AM, CM) (graph A–M–C), C is conditionally independent of A, so the AM association is collapsible over C: fitted conditional and marginal AM odds ratios both equal 61.9 (Example 8.8); similarly CM (25.1). The AC association is not collapsible, because M is linked to both A and C: conditional AC odds ratio 1.0, marginal 2.7. In (AC, AM, CM) no collapsibility condition holds, and every pair has quite different conditional and marginal associations.

9.1.3 Collapsibility and logit models

In a multicenter clinical trial with binary treatment x, binary response Y and centers Z, the model $\operatorname{logit} P(Y = 1 \mid X = i, Z = k) = \alpha + \beta x_i + \beta_k^Z$ corresponds to (XY, XZ, YZ), so the common conditional odds ratio e^{β} typically differs from the odds ratio of the marginal 2×2 table. The simpler model without center effects, $\operatorname{logit} = \alpha + \beta x_i$, states $Y \perp Z \mid X$; it is equivalent to (XY, XZ), which is collapsible over Z. So when center effects are negligible the treatment effect is approximately the marginal odds ratio.

9.1.4 Collapsibility and association graphs for multiway tables

Theorem 9.4 (Parametric collapsibility, Bishop et al.)

Partition the variables into disjoint sets A, B, C such that B separates A and C in the association graph. After collapsing the table over the variables in C, the parameters relating variables within A, and those relating variables in A to variables in B, are unchanged.

Proof. [Supplement]

Step 1. No term contains variables from both A and C (that would create an A–C edge, a path avoiding B). Put all terms meeting A into f and all others into g; then each term of f lies in $A \cup B$ and each term of g lies in $B \cup C$:

$$\mu(\mathbf{a}, \mathbf{b}, \mathbf{c}) = f(\mathbf{a}, \mathbf{b})\, g(\mathbf{b}, \mathbf{c}), \qquad \log f = \sum_{T \cap A \neq \emptyset} \lambda^T.$$

Step 2. Collapsing over C: $\mu(\mathbf{a}, \mathbf{b}) = f(\mathbf{a}, \mathbf{b})\, G(\mathbf{b})$ with $G(\mathbf{b}) = \sum_{\mathbf{c}} g(\mathbf{b}, \mathbf{c})$. Thus

$$\log \mu(\mathbf{a}, \mathbf{b}) = \sum_{T \cap A \neq \emptyset} \lambda^T + \log G(\mathbf{b}).$$

Step 3. $\log G(\mathbf{b})$ decomposes into terms involving only variables of B. Since the decomposition of $\log \mu(\mathbf{a}, \mathbf{b})$ into main-effect and interaction terms is unique under fixed (e.g. baseline) constraints, every term involving a variable of A in the collapsed table equals the corresponding λ^T of the full model. ■

For (WX, WY, WZ, YZ): with $A = \{X\}$, $B = \{W\}$, $C = \{Y, Z\}$, the WX association is unchanged after collapsing over Y, Z; with $A = \{Y, Z\}$, $B = \{W\}$, $C = \{X\}$, the associations among W, Y, Z are unchanged after collapsing over X. A variable independent of all others can be collapsed over without affecting other terms. When B has several variables the parameters are unchanged but their ML estimates may differ slightly; estimates also agree ("commutativity of fitting and collapsing") if the model contains the highest-order term relating the variables of B to each other.

9.2 Model selection and comparison

Motivation. Even with five variables there are thousands of hierarchical loglinear models. We need principles (which terms must be included) and tools (likelihood-ratio comparisons of nested models) to find a model that fits well but is simple enough to interpret.

9.2.1 Considerations in model selection

A model should be complex enough to fit well yet simple to interpret, smoothing rather than overfitting. Guidelines:

(i) Distinguish responses from explanatory variables and concentrate on terms linking responses and linking explanatory variables to responses.

(ii) Include the most general interaction among the explanatory variables, and any margin fixed by the sampling design; then the likelihood equations reproduce these totals. For Table 8.8 with responses I, S and predictors G, L, the model should contain GL (so the fit has 20,629 urban women as observed), i.e. be at least as complex as (GL, S, I).

(iii) With a single categorical response, use the equivalent logit model.

(iv) For exploration, fit the model with all single-factor terms, then all two-factor terms, then all three-factor terms, and so on, to locate the needed complexity; automatic searches (backward elimination) should be used with care.

9.2.2 Model building for the Dayton student survey

Example 9.5 (Five-way student survey)

The survey of Example 8.8 also recorded gender (G) and race (R):

		Race = white				Race = other			
		Female		Male		Female		Male	
A	C	M=yes	no	yes	no	yes	no	yes	no
yes	yes	405	268	453	228	23	23	30	19
yes	no	13	218	28	201	2	19	1	18
no	yes	1	17	1	17	0	1	1	8
no	no	1	117	1	133	0	12	0	17

Treating A, C, M as responses and G, R as explanatory, every model contains GR.

Model	G^2	df
1. Mutual independence $+ GR$	1325.1	25
2. Homogeneous association (all two-factor terms)	15.3	16
3. All three-factor terms	5.3	6
4g. (2) $- CR$	15.8	17
5. $(AC, AM, CM, AG, AR, GM, GR, MR)$	16.7	18
6. $(AC, AM, CM, AG, AR, GM, GR)$	19.9	19
7. (AC, AM, CM, AG, AR, GR)	28.8	20

Model 3 improves on model 2 only by $15.3 - 5.3 = 10.0$ on 10 df, so three-factor terms are dropped. Backward elimination from model 2 (removing each of the 9 candidate pairs, models 4a–4i, e.g. removing AC gives 201.2 and removing CM gives 513.5) first removes CR (increase 0.5, df 1), then CG (+0.9), then MR (+3.2); removing AG next would add 5.3 ($P = 0.02$), so we stop at model 6. (These P-values are not literal, since the data chose the tests; many counts are small, but G^2 differences remain useful.)

In model 6 (Figure 1, middle) every path from C to $\{G, R\}$ goes through $\{A, M\}$: given alcohol and marijuana use, cigarette use is independent of gender and race. By Theorem 9.4, the AC and CM conditional associations equal those of (AC, AM, CM) fitted to the ACM table collapsed over G, R. Model 7 drops GM; then A separates $\{G, R\}$ from $\{C, M\}$, so all associations among A, C, M equal those of (AC, AM, CM) in the collapsed table. Model 7 fits reasonably ($G^2 = 28.8$, df $= 20$, $\hat{\Delta} = 0.036$), so one may collapse over gender and race to study A, C, M; the full model additionally estimates the effects of G and R on the responses.

9.2.3 Loglinear model comparison statistics

Let M_0 be a special case of M_1, with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$.

Theorem 9.6 (Partitioning G^2)

$$G^2(M_0) = G^2(M_1) + G^2(M_0 \mid M_1), \tag{9.1}$$

where the likelihood-ratio statistic for M_0 against M_1 has the form

$$G^2(M_0 \mid M_1) = G^2(M_0) - G^2(M_1) = 2\sum_i \hat{\mu}_{1i} \log \frac{\hat{\mu}_{1i}}{\hat{\mu}_{0i}}, \tag{9.2}$$

which is the G^2 statistic for M_0 with $\hat{\boldsymbol{\mu}}_1$ playing the role of the data.

Proof.

Step 1. $G^2(M) = 2\sum n_i \log(n_i/\hat{\mu}_i)$, so

$$G^2(M_0) - G^2(M_1) = 2\sum_i n_i(\log \hat{\mu}_{1i} - \log \hat{\mu}_{0i}).$$

Step 2. Write $\log \hat{\boldsymbol{\mu}}_1 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1$ and, since $M_0 \subseteq M_1$, $\log \hat{\boldsymbol{\mu}}_0 = \mathbf{X}_0\hat{\boldsymbol{\beta}}_0 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_0^*$, where $\hat{\boldsymbol{\beta}}_0^*$ is $\hat{\boldsymbol{\beta}}_0$ with zeros appended for the extra parameters. Then

$$2\mathbf{n}'(\log \hat{\boldsymbol{\mu}}_1 - \log \hat{\boldsymbol{\mu}}_0) = 2\,\mathbf{n}'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*) = 2\,\hat{\boldsymbol{\mu}}_1'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*),$$

using the likelihood equations $\mathbf{X}_1'\mathbf{n} = \mathbf{X}_1'\hat{\boldsymbol{\mu}}_1$ for M_1 (Theorem 8.18). This is $2\sum \hat{\mu}_{1i}\log(\hat{\mu}_{1i}/\hat{\mu}_{0i})$.

Step 3. $G^2(M_0 \mid M_1)$ has the G^2 form. (Note also $\sum \hat{\mu}_{0i} = \sum \hat{\mu}_{1i} = n$ when both models contain an intercept.) Equation (9.1) is the definition rearranged. ■

$G^2(M_0 \mid M_1)$ depends on the data only through $\hat{\boldsymbol{\mu}}_1$, i.e. through the sufficient statistics of M_1. The Pearson difference $X^2(M_0) - X^2(M_1)$ need not be nonnegative; the natural Pearson analogue is

$$X^2(M_0 \mid M_1) = \sum_i \frac{(\hat{\mu}_{1i} - \hat{\mu}_{0i})^2}{\hat{\mu}_{0i}}. \tag{9.3}$$

When M_0 holds, both are asymptotically χ^2 with $\mathrm{df} = \mathrm{df}(M_0) - \mathrm{df}(M_1)$, and they behave similarly even in fairly sparse tables (Haberman). When M_1 holds but M_0 does not, $G^2(M_1)$ remains χ^2 while the others grow without bound.

9.2.4 Partitioning chi-squared with model comparisons

Applying (9.1) repeatedly to a nested sequence $M_1 \supset M_2 \supset \cdots$ partitions G^2 into components that are asymptotically independent. Example: for a $2 \times J$ table let M_j state that columns $1, \ldots, j$ form an independence table. Then

$$G^2(M_J) = G^2(M_2) + G^2(M_3 \mid M_2) + \cdots + G^2(M_J \mid M_{J-1}),$$

and substituting the fitted values shows that $G^2(M_j \mid M_{j-1})$ is the G^2 for independence in the 2×2 table whose first column combines columns $1, \ldots, j-1$ and whose second column is column j. This justifies the partitioning of Section 3.3.3.

With s planned tests for nested models, using level $1-(1-\alpha)^{1/s}$ for each keeps the overall asymptotic type I error at most α (the tests being asymptotically independent). For $s = 3$ and $\alpha = 0.05$, use $1 - 0.95^{1/3} = 0.017$ per test.

9.2.5 Identical marginal and conditional tests of independence

When both models are direct, they contain the independence linkages that guarantee collapsibility, and the conditional test reduces to a marginal one (Sundberg, 1975): if two direct models differ

only by one pairwise term, $G^2(M_0 \mid M_1)$ equals G^2 for independence in the marginal table of that pair.

Proposition 9.7 (A conditional test that equals a marginal test)

$G^2[(X,Y,Z) \mid (XY,Z)]$ equals $G^2(X,Y)$ for testing independence in the marginal XY table.

Proof. By Theorem 8.20, $\hat{\mu}_{1,ijk} = n_{ij+}n_{++k}/n$ and $\hat{\mu}_{0,ijk} = n_{i++}n_{+j+}n_{++k}/n^2$, so $\hat{\mu}_1/\hat{\mu}_0 = n\,n_{ij+}/(n_{i++}n_{+j+})$ does not depend on k. By (9.2),

$$G^2 = 2\sum_{i,j,k} \frac{n_{ij+}n_{++k}}{n} \log \frac{n\,n_{ij+}}{n_{i++}n_{+j+}} = 2\sum_{i,j} n_{ij+} \log \frac{n_{ij+}}{n_{i++}n_{+j+}/n},$$

after summing n_{++k}/n over k. This is G^2 for independence in the $\{n_{ij+}\}$ table. ■

This is expected: under (XY,Z) the XY association is collapsible over Z.

9.3 Diagnostics for checking models

Motivation. G^2 tells us whether a model fits; residuals tell us where it fails, cell by cell.

9.3.1 Residuals for loglinear models

Definition (Pearson and standardized residuals)

For cell i the *Pearson residual* is

$$e_i = \frac{n_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}, \qquad \sum_i e_i^2 = X^2. \tag{9.4}$$

The *standardized Pearson residual* is $r_i = e_i/\sqrt{1-\hat{h}_i}$, where $\hat{h}_i$ is the leverage (diagonal of the estimated hat matrix).

Pearson residuals have asymptotic variances below 1 (on average residual df/number of cells); r_i is asymptotically $N(0,1)$ and is preferable. Deviance residuals are an alternative.

9.3.2 Student survey example revisited

For the five-way survey with all two-factor terms, the only large standardized residual is 3.2 (fitted 3.1 for a count of 8); for model 6 it is 3.3 (fitted 2.9) in the same cell: nonwhite males who smoked cigarettes but used neither alcohol nor marijuana. Given the many cells and large n, neither model looks problematic.

9.3.3 Correspondence between loglinear and logit residuals

Proposition 9.8 (Logit versus loglinear Pearson residuals)

For equivalent logit and loglinear models with binary response, the logit-model Pearson residual for a success count equals the loglinear Pearson residual for that cell divided by $\sqrt{1-\hat{\pi}}$; hence it is larger in absolute value.

Proof. Let the cell have count y out of N trials in its binomial row. The fitted values agree: $\hat{\mu} = N\hat{\pi}$ (Theorem 8.16). The logit residual is $(y - N\hat{\pi})/\sqrt{N\hat{\pi}(1-\hat{\pi})}$ and the loglinear residual is $(y-\hat{\mu})/\sqrt{\hat{\mu}} = (y - N\hat{\pi})/\sqrt{N\hat{\pi}}$. Their ratio is $1/\sqrt{1-\hat{\pi}} > 1$. ■

After standardization by their own estimated standard errors, the two kinds of residual are identical, another reason to prefer standardized residuals.

9.4 Modeling ordinal associations

Motivation. Nominal loglinear models ignore category order: permuting rows gives the same fit. For two ordinal variables the saturated model is useless and independence often fails mostly in the corner cells; a single "trend" parameter captures this.

Example 9.9 (Premarital sex and teenage birth control)

1991 General Social Survey. Rows: premarital sex is always wrong (1), almost always wrong (2), wrong only sometimes (3), not wrong at all (4). Columns: teenage birth control should be available, strongly disagree (1) to strongly agree (4). Entries: count, (independence fit), [standardized residual], {L×L fit}.

	SD	D	A	SA
1	81 (42.4) [7.6] {80.9}	68 (51.2) [3.1] {67.6}	60 (86.4) [−4.1] {69.4}	38 (67.0) [−4.8] {29.1}
2	24 (16.0) [2.3] {20.8}	26 (19.3) [1.8] {23.1}	29 (32.5) [−0.8] {31.5}	14 (25.2) [−2.8] {17.6}
3	18 (30.1) [−2.7] {24.4}	41 (36.3) [1.0] {36.1}	74 (61.2) [2.2] {65.7}	42 (47.4) [−1.0] {48.8}
4	36 (70.6) [−6.1] {33.0}	57 (85.2) [−4.6] {65.1}	161 (143.8) [2.4] {157.4}	157 (111.4) [6.8] {155.5}

Independence fits poorly, $G^2(I) = 127.6$, df $= 9$; the corner residuals show a positive trend.

9.4.1 Linear-by-linear association in two-way tables

Definition (Linear-by-linear association (L×L) model)

With ordered row scores $u_1 \le \cdots \le u_I$ and column scores $v_1 \le \cdots \le v_J$,

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j, \tag{9.5}$$

the special case $\lambda_{ij}^{XY} = \beta u_i v_j$ of the saturated model: one association parameter instead of $(I-1)(J-1)$. Independence is $\beta = 0$. With $u_i = i$, $v_j = j$ it is the *uniform association* model.

Theorem 9.10 (Odds ratios under L×L)

Under (9.5), for rows $a < c$ and columns $b < d$,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \beta(u_c - u_a)(v_d - v_b). \tag{9.6}$$

With unit-spaced scores all local odds ratios equal e^{β}. If $\beta > 0$, then Y tends to increase with X and μ_{ij} exceeds its independence value where X, Y are both high or both low.

Proof.

Step 1. By Theorem 8.3(b) the log odds ratio equals $\lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}$ with $\lambda_{ij}^{XY} = \beta u_i v_j$:

$$\beta(u_a v_b + u_c v_d - u_a v_d - u_c v_b) = \beta(u_c - u_a)(v_d - v_b).$$

Step 2. For adjacent rows and columns with unit spacing, $(u_{i+1} - u_i)(v_{j+1} - v_j) = 1$.

Step 3. For $\beta > 0$ all odds ratios with $a < c$, $b < d$ exceed 1, which is positive (likelihood-ratio type) dependence. The deviation $\beta u_i v_j$ from independence is largest in absolute value at the corners. ■

The deviation $\beta u_i v_j$ is linear in the column scores within a row (slope βu_i) and linear in the row scores within a column, hence the name. Scores are often midpoints of an underlying continuous scale, or standardized (mean 0, variance 1 under the margins); then β is a log odds ratio for unit standard-deviation changes, and for an underlying bivariate normal with correlation ρ, $\beta \approx \rho/(1-\rho^2)$ ($\approx \rho$ for weak association).

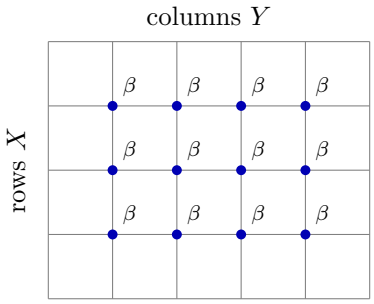


Figure 2: Uniform association: every 2×2 table of adjacent rows and columns has log odds ratio β (book Figure 9.2).

9.4.2 Corresponding logit model for adjacent responses

Proposition 9.11 (Adjacent-categories logit form)

Let $\pi_{j|i} = P(Y = j \mid X = i)$. Under (9.5),

$$\log \frac{\pi_{j+1|i}}{\pi_{j|i}} = (\lambda_{j+1}^{Y} - \lambda_{j}^{Y}) + \beta(v_{j+1} - v_j)u_i,$$

and for unit-spaced $\{v_j\}$ this is $\alpha_j + \beta u_i$: the same linear effect of X on all $J-1$ adjacent-categories logits.

Proof. $\pi_{j+1|i}/\pi_{j|i} = \mu_{i,j+1}/\mu_{ij}$; subtracting (9.5) at j from (9.5) at $j+1$, λ and λ_i^X cancel, leaving the stated expression. ■

9.4.3 Likelihood equations and model fitting

Theorem 9.12 (Likelihood equations for L×L)

The ML fit of (9.5) satisfies

$$\hat{\mu}_{i+} = n_{i+}, \qquad \hat{\mu}_{+j} = n_{+j}, \qquad \sum_{i,j} u_i v_j \hat{\mu}_{ij} = \sum_{i,j} u_i v_j n_{ij}.$$

Hence the fitted and observed distributions have the same marginal means and variances of the scores and the same correlation between them. The residual df are $IJ - I - J$.

Proof.

Step 1. The Poisson log likelihood is

$$L = n\lambda + \sum_i n_{i+}\lambda_i^X + \sum_j n_{+j}\lambda_j^Y + \beta \sum_{i,j} u_i v_j n_{ij} - \sum_{i,j} \exp(\lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j).$$

Differentiating with respect to λ_i^X, λ_j^Y and β and equating to zero gives the three sets of equations (the sufficient statistics are the margins and $\sum u_i v_j n_{ij}$).

Step 2. Dividing by n, the fitted proportions $\hat{\pi}_{ij}$ and the sample proportions p_{ij} have equal margins, hence equal means and variances of u and of v, and equal $\sum u_i v_j \hat{\pi}_{ij} = \sum u_i v_j p_{ij}$, hence equal covariance and correlation.

Step 3. The model has $1 + (I-1) + (J-1) + 1$ parameters, so df $= IJ - (I + J)$, unsaturated except for 2×2 tables. ■

The fitted counts therefore reproduce the trend of the data. Fitting uses Newton–Raphson.

9.4.4 Sex opinions example

Example 9.13 (L×L fit to Example 9.9)

With scores $1, 2, 3, 4$ for both variables, $G^2(\text{L}\times\text{L}) = 11.5$, df $= 8$, a dramatic improvement over 127.6 on 9 df, especially in the corners. $\hat{\beta} = 0.286$ (SE 0.028): local odds ratio $e^{0.286} = 1.33$, 95% CI $\exp(0.286 \pm 1.96 \times 0.028) = (1.26, 1.41)$. Nonlocal odds ratios are larger: for the four corner cells, by (9.6),

$$\exp[0.286(4-1)(4-1)] = 13.1 = \frac{80.9 \times 155.5}{29.1 \times 33.0}.$$

Rescaled equally spaced scores (e.g. rows $2, 4, 6, 8$) give the same fit with $\hat{\beta}$ rescaled (0.143). Scores $1, 2, 4, 5$ (treating categories 2 and 3 as farther apart) give $G^2 = 8.8$, $\hat{\beta} = 0.146$. Standardized scores give $\hat{\beta} = 0.374$, and solving $\beta = \rho/(1-\rho^2)$ gives $\hat{\rho} = 0.333$ for an underlying bivariate normal.

9.4.5 Directed ordinal test of independence

Within L×L, independence is $H_0 : \beta = 0$, tested by

$$G^2(I \mid \text{L} \times \text{L}) = G^2(I) - G^2(\text{L} \times \text{L}), \qquad \text{df} = 1.$$

For Example 9.9: $127.6 - 11.5 = 116.1$; the Wald statistic $(\hat{\beta}/\text{SE})^2 = 102.5$; the score statistic is the correlation statistic $M^2 = (n-1)r^2 = 112.6$ of Section 3.4.1.

Why the ordinal test is more powerful

When L×L holds, $G^2(I \mid \mathrm{L}\times\mathrm{L})$ and $G^2(I)$ have (asymptotically) the same noncentrality, but the former has df $= 1$ instead of $(I-1)(J-1)$. For a fixed noncentrality, the power of a χ^2 test decreases as df increase, so the directed test is more powerful, and the advantage grows with I and J.

9.5 Poisson regression for rates

Motivation. When events are counted over different amounts of exposure (time, space, population), the rate, not the raw count, is the natural response. A loglinear model with an offset models the rate while keeping the Poisson count as the observation.

9.5.1 Analyzing rates using loglinear models with offsets

Proposition 9.14 (Rate model with offset)

For counts n_i with exposure (index) t_i, the model

$$\log(\mu_i/t_i) = \alpha + \beta x_i \tag{9.7}$$

is the Poisson loglinear model $\log \mu_i = \log t_i + \alpha + \beta x_i$, in which $\log t_i$ is an offset (a predictor whose coefficient is fixed at 1). Equivalently $\mu_i = t_i \exp(\alpha + \beta x_i)$: the mean is proportional to the exposure, and e^β is the multiplicative effect of a unit increase in x on the rate.

Proof. Add $\log t_i$ to both sides of (9.7) and exponentiate. The ratio of rates at $x+1$ and x is $e^{\alpha+\beta(x+1)}/e^{\alpha+\beta x} = e^\beta$. ■

The identity link $\mu_i/t_i = \alpha + \beta x_i$, i.e. $\mu_i = \alpha t_i + \beta x_i t_i$, needs no offset: it is a Poisson GLM with identity link, predictors t_i and $x_i t_i$ and no intercept. Effects are additive rather than multiplicative; fitting may fail if negative fitted values arise.

9.5.2 Modeling death rates for heart valve operations

Example 9.15 (Heart valve replacement)

109 patients, classified by valve type and age; time at risk is the total follow-up (months) of all patients in the cell, those who died and those censored.

Age		Aortic	Mitral
< 55	Deaths	4	1
	Time at risk	1259	2082
	Death rate	0.0032	0.0005
55+	Deaths	7	9
	Time at risk	1417	1647
	Death rate	0.0049	0.0055

A binary model for the $2 \times 2 \times 2$ table (died yes/no) would be wrong, since patients had different exposures (3 to 97 months). With dummies a (age 55+) and v (mitral),

$$\log(\mu_{ij}/t_{ij}) = \alpha + \beta_1 a_i + \beta_2 v_j. \tag{9.8}$$

Estimates: $\hat{\beta}_1 = 1.221$ (SE 0.514), $\hat{\beta}_2 = -0.330$ (SE 0.438). Given valve type, the older group's death rate is $e^{1.221} = 3.4$ times the younger group's; 95% Wald CI $\exp(1.221 \pm 1.96 \times 0.514) = (1.2, 9.3)$, likelihood-ratio CI $(1.3, 10.4)$. Fit: $G^2 = 3.2$, $X^2 = 3.1$, df $= 1$; the model without valve type has $G^2 = 3.8$ (df 2); models without age fit poorly. Fitted death rates: < 55: 0.0018 (aortic), 0.0013 (mitral); 55+: 0.0062, 0.0044. The identity-link model fits with $G^2 = 1.1$ (df 1); its $\hat{\beta}_1 = 0.0040$ (SE 0.0014) is an additive difference in death rates. Only 21 of 109 patients died, so estimates are imprecise, but every patient contributes through the times at risk.